Biomedical Signal Processing and Control 91 (2024) 106016

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

An atrial fibrillation detection algorithm based on lightweight design architecture and feature fusion strategy

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ARTICLE INFO

Keywords:
Atrial fibrillation
Electrocardiogram
Deep learning
Lightweight
Attention mechanism
Cross-guidance mechanism

ABSTRACT

Background: Atrial fibrillation (AF) is one of the common types of cardiac arrhythmias, and its medical burden is continuously increasing. Wearable ECG signal analysis based on deep learning (DL) is an effective approach for screening AF. However, existing DL algorithms require extensive computational resources for AF recognition, hindering their clinical applicability.
Objective: This study aims to develop a lightweight DL model to address the challenges of DL algorithms in the clinical AF recognition domain.
Method: Using a distributed approach, layer-by-layer cross-guidance mechanism, and attention fusion mechanism, we designed a lightweight cross-guidance network (LCG-Net). The main path uses lightweight depth-wise separable convolutions to extract deep-level information of AF, while the auxiliary path uses standard convolutions to compensate for the weak feature expression capability of depth-wise separable convolutions. Based on the idea of mutual guidance, a layer-by-layer cross-guidance mechanism is designed to achieve information interaction and fusion between depthwise separable convolutions and standard convolutions. An attention fusion mechanism is developed based on attention mechanism and 2D convolution templates to select and precisely fuse information from different paths and different layers.
Result: LCG-Net has only 39.04 K parameters and 8.16 M computations. On a clinical dataset consisting of ECG records from 252 patients, it achieved accuracy and F1 scores of 98.39 % and 98.38 %, respectively.
Conclusion: The proposed LCG-Net demonstrates excellent lightweight, stability, and accuracy, holding promising prospects in the clinical diagnosis of AF.

1. Introduction

Atrial fibrillation (AF) is one of the most common types of arrhythmia, which increases the risk of stroke by 5 times, and the risk of dementia and myocardial infarction by 2 times [1]. In particular, paroxysmal AF has paroxysmal and sporadic characteristics, which are difficult to detect using short-term clinical ECG. 90 % of patients show no obvious symptoms, and only a few patients are detected and enter the clinical intervention stage [2]. The low detection rate has always been a major problem in the treatment of AF [3]. In order to achieve early detection and intervention of AF, and reduce the mortality rate of patients, wearable long-term ECG monitoring devices (such as Holter) have been widely used in clinical practice. But, existing DL learning algorithms require powerful hardware, such as CPUs and GPUs on PCs, which are usually bulky and difficult to integrate into wearable devices, making it difficult to integrate DL algorithms into existing wearable devices. Additionally, smartwatches with AF detection functions are widely used worldwide, studies have shown [4] that the quality of signals and detection accuracy of existing smartwatches have not reached clinical standards. Developing a lightweight AF detection algorithm that can be embedded in low-end chips and integrated into clinical wearable ECG monitors is the key to solving the problem.

Traditional AF detection algorithms extract features using prior knowledge and classify them using machine learning models. For example, Abhimanyu [5], Sun [6], Shadnaz [7], and Szymon et al. [8] extracted time–frequency features of RR intervals, systolic peaks,

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https://doi.org/10.1016/j.bspc.2024.106016
Received 24 August 2023; Received in revised form 23 December 2023; Accepted 29 January 2024
Available online 6 February 2024
1746-8094/

logarithmic energy entropy, and HRV parameters, respectively. Furthermore, Krasteva et al. [9], Rouhi et al. [10] and Christov et al. [11] used feature importance ranking algorithms to optimize the obtained features. These algorithms have made significant progress in AF detection. However, with deep learning demonstrating strong feature extraction capabilities in various fields, an increasing number of researchers are applying it to AF detection and achieving performance superior to traditional machine learning methods [12].

Researchers have conducted in-depth studies on the depth, width, and input format of the models. For example, Zhang et al. [13] developed a multi-path DL model based on the feature reuse principle of Dense Net and the memory function of Long Short-Term Memory (LSTM). However, each path uses 30 layers of convolution and 4 layers of LSTM, resulting in a significant increase in computational cost. Zhang et al. [14] proposed a multi-branch convolutional neural network and a Bi-LSTM network, with each branch consisting of 12 layers of convolution. However, the number of branches in the network is dynamically changing, and the highest can produce 24 branches, resulting in much higher actual computational cost than the literature [13]. Eedara et al. [15] established a multi-task deep convolutional neural network (MT-DCNN) composed of 11 layers of convolution and 11 layers of deconvolution based on the idea of auxiliary classifiers in Google Net. MT-DCNN reconstructs ECG as an auxiliary task to improve the robustness of the network in AF recognition. However, with the increase in task complexity, the complexity and training difficulty of the network have both increased significantly. Zachi et al. [16] input the 12-lead ECG into a network composed of 10 residual modules to obtain complete ECG information. However, this algorithm applied a total of 120 residual modules, resulting in huge computational cost and low real-time performance. Some researchers [17,18,19,20] have transformed one-dimensional signals into two-dimensional images and matrix. However, compared to analyzing one-dimensional signals, DL models require higher computational costs when analyzing two-dimensional images and matrix. For example, converting a 5-second time segment sampled at 100 Hz into a time–frequency image with a size of 500×61 using wavelet transformation increases the computational load by a factor of 61. The above algorithms increase the complexity of network structure and input samples while improving the accuracy of AF detection, resulting in a significant increase in computational cost, which exceeds the computational requirements of low-end chips (such as FPGA) and cannot be embedded in existing clinical wearable ECG devices. As a result, ECG analysis tasks in clinical practice can only be performed by ECG experts, reducing the real-time performance of AF screening.

Researchers have reduced the computational cost of algorithms by compressing inputs and simplifying structures. For example, Zhang et al. [21], Lai et al. [22], Liu et al. [23], and Vykintas et al. [24] trained neural networks using artificially extracted features such as RR sequences or heart rate variability to reduce computational costs by simplifying inputs. However, due to the limitations of prior knowledge and subjectivity, artificially extracted features are difficult to cover all the information in ECG. Yu et al. [25] downsampled the raw ECG sequence to speed up the inference speed of the algorithm by shortening the length of the input signal, but this method loses the original information in ECG. From an information perspective, DL requires sufficient information as a driver, and using artificially extracted features or downsampled signals as inputs will lose the implicit information in ECG, resulting in insufficient training of DL models. While ensuring the completeness of input information, some researchers use raw ECG as input and use parameter quantization technique [26] to reduce the inference cost of algorithms. However, this technique loses some information during feature extraction and causes a decrease in algorithm accuracy. Another group of researchers directly generalize lightweight models in the field of computer vision (such as CSP Net [27], Squeeze Net [28], and Mix Net [29]) to AF recognition tasks [30]. However, frameworks based on two-dimensional convolution templates focus on spatial information in feature extraction, and their ability to express temporal features in one-dimensional signals is limited, resulting in ECG features lacking temporal resolution. Moreover, the generality of artificial intelligence algorithms between different fields is controversial. When two-dimensional convolution models are directly generalized to one-dimensional convolution models, their generality and stability in new tasks need further exploration.

This study aims to develop a lightweight cross-guidance network (LCG-Net) to address three challenges faced by existing AF identification algorithms.

- Implementing a distributed network architecture to achieve lightweight design of DL models, addressing the significant increase in computational costs when existing DL models pursue high accuracy and generalization.
- Utilizing a layer-by-layer cross-guidance mechanism to facilitate information interaction among multiple pathways, addressing the issue of obstructed information flow in distributed networks.
- Using attention fusion mechanism to relearn information from different sources, solving the problem of redundant information in feature maps output from multiple pathways.

2. Material and method

2.1. Material

The MIT-BIH Atrial Fibrillation Database (AFDB) [31] comprises 23 ECG records, with records 0735 and 0366 lacking annotations. All records are sourced from patients with AF (primarily paroxysmal), and the sampling frequency is set at 250 Hz. AFDB contains four types of rhythm annotations: atrial fibrillation, atrial flutter, junctional rhythm, and normal. According to clinical expert recommendations, junctional rhythm and normal are categorized as non-atrial fibrillation rhythms (N), while atrial fibrillation and atrial flutter are categorized as atrial fibrillation rhythms (AF). In this study, the sampling rate of AFDB was standardized to 200 Hz.

To evaluate the practical application of the proposed method, we established a clinical dataset called the Shandong Provincial Hospital Database (SPHDB). The SPHDB contains 24-hour wearable ECG recordings from 250 patients, all of whom were diagnosed with paroxysmal atrial fibrillation. The inclusion criteria for patients are those who experienced AF events during a 24-hour period while wearing a cardiac monitor correctly between 2022 and 2023. Some patients may have noise interference and concurrent multiple diseases. The patients were from different regions, with a minimum age of 42 and a maximum age of 90, and the male-to-female ratio was approximately 1:1. All recordings were collected using a 12-lead Holter wearable ECG monitor with a sampling frequency of 200 Hz. During the data collection process, patients were in a completely unrestricted state of activity, and the data collection process procedures did not interfere with their normal daily life within a 24-hour period. Each record was annotated by two physicians, and a senior physician was responsible for reviewing the results from both annotators. The annotated data was categorized into AF and N classes. This protocol was approved by the Ethics Committee of Shandong Provincial Hospital.

At the same time, this study constructed an augmented dataset (AD), which includes two patients with paroxysmal atrial fibrillation. Unlike the 250 patients in the SPHDB, both patients in the AD had 24-hour lead II ECG recordings showing unstable baselines or baseline disappearance after the T wave. The AD was included in the training set of SPHDB to create SPHDB Plus. In SPHDB Plus, any information from the AD will not appear in the validation set or test set.

In this study, we used the ECG recordings of lead II as the research object. Firstly, we used the db6 wavelet to perform 9-level decomposition on all ECG signals to remove high-frequency noise and low-frequency noise [14]. Then, the obtained signals were segmented into 15-second ECG segments [61]. Each 15-second segment is labeled as either N or AF. If atrial fibrillation occurs in any 15-second segment, the label is AF. The label is set as N only when no atrial fibrillation occurs in the 15-second segment. Finally, we standardized the segmented ECG segments using Z-score normalization to speed up the gradient descent process for optimal solution.

After preprocessing, this study obtained 32,815 N samples and 20,685 AF samples from the AFDB, totaling 53,500 samples. We divided all samples into training, validation, and test sets using intra-patient and stratified sampling, with proportions of 80 %, 10 %, and 10 %, respectively.

After preprocessing, this study obtained 125,000 N samples and 125,000 AF samples from the SPHDB, totaling 250,000 samples. We divided the 250,000 samples into 5 groups using an inter-patient paradigm, labeled as G1, G2, G3, G4, and G5, each containing 50,000 samples (from 50 patients). One group was selected for training (35 patients) and validation (15 patients), while the remaining four groups (50 patients × 4) were used for testing, conducting a total of 5 experiments. The division of training, validation, and test sets for each experiment is shown in Fig. 1.

2.2. Method

LCG-Net uses a distributed approach to design two lightweight paths, combined with a layer-by-layer cross-guidance mechanism and an attention fusion mechanism to achieve feature extraction, feature selection, and feature fusion. The overall structure of the network is shown in Fig. 2, where SC represents standard convolution and DC represents depth separable convolution. The main path is constructed by 9 layers of 1 × 3 × 32 (32 convolutional kernels of size 1 × 3) DC, and the auxiliary path is constructed by 9 layers of 1 × 3 × 16 (16 convolutional kernels of size 1 × 3) standard convolution. The layer-by-layer cross-guidance mechanism is composed of 2 sets of cross-multiplying attention modules, and the attention fusion mechanism is constructed by an attention mechanism and 1 layer of 3 × 1 × 32 (32 convolutional kernels of size 3 × 1) two-dimensional convolution template.

2.2.1. Distributed network structure

In the early development of neural networks, the computing power of CPUs/GPUs limited the depth and width of network development. In order to achieve deeper networks, Alex Net [32] innovatively applied a distributed approach, splitting a large network into two small networks and using dual GPUs to load convolution kernels separately. However, communication between different GPUs was only possible at specific layers, which greatly hindered the flow of information. Subsequently, the field of computer vision conducted in-depth research on lightweight models. Squeeze Net applied the distributed approach of Alex Net to each layer, using a large number of 1 × 1 convolution kernels instead of 3 × 3 convolution kernels, with 1 × 1 convolution kernels used to increase the number of feature maps and 3 × 3 convolution kernels used to extract features. Squeeze Net reduced the computational cost of Alex Net without reducing accuracy. During the same period, Andrew et al. [33], Zhang et al. [34] and Chollet et al. [35] respectively proposed Mobile Net, Shuffle Net, and Xception Net, which enabled DL models to move towards lightweight development while ensuring high accuracy. In the field of computer vision, most lightweight models are built based on DC or its variants. DC applies the idea of channel-wise convolution, where each convolution kernel only convolves with one feature map, and each feature map can only be convolved by one convolution kernel. Compared with standard convolution, DC has fewer parameters and computations, resulting in higher computational efficiency. The difference between standard convolution and DC is shown in Fig. 3.

The parametric quantity P_D and the calculated quantity F_D of the DC calculation process are shown in Eqs. (1) and (2):

$$P_D = K_D \times I_D \tag{1}$$

$$F_D = K_D \times (L - K_D + 1) \times I_D \tag{2}$$

K_D represents the size of the convolutional kernel for DC, I_D represents the number of input channels for DC, and L represents the length of the input signal.

The parameter count P_S and the computational cost F_S for standard convolution are given by the Eqs. (3) and (4):

$$P_S = \mathrm{I} \times K \times C \tag{3}$$

$$F_S = \mathrm{I} \times K \times (L - K + 1) \times C \tag{4}$$

K represents the kernel size of standard convolution, C represents the number of kernels in standard convolution, and I represents the number of input channels. When DC is consistent with the hyperparameters of standard convolution, that is, $K_D = K$ and $I_D = I$,

$$\frac{P_D}{P_S} = \frac{K_D \times I_D}{I \times K \times C} = \frac{1}{C} \tag{5}$$

$$\frac{F_D}{F_S} = \frac{K_D \times (L - K_D + 1) \times I_D}{I \times K \times (L - K + 1) \times C} = \frac{1}{C} \tag{6}$$

From Eqs. (5) and (6), it can be seen that when extracting features from the same input, the parameter and computational costs of DC are only $1/C$ of standard convolution.

In this study, the distributed approach and DC were combined to construct two complementary paths. The main path is constructed using DC to ensure that features are acquired at low computational cost. Considering that ECG is a time-series signal, different features maintain

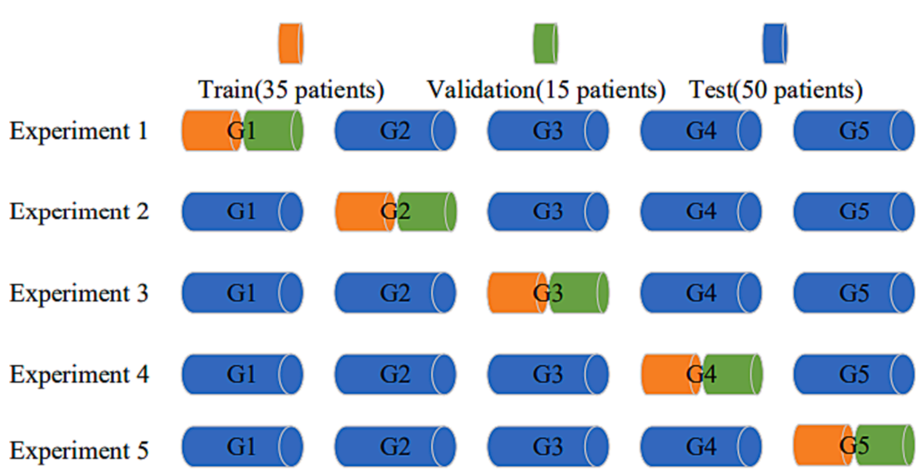

Fig. 1. Division of training, validation and test sets for SPHDB.

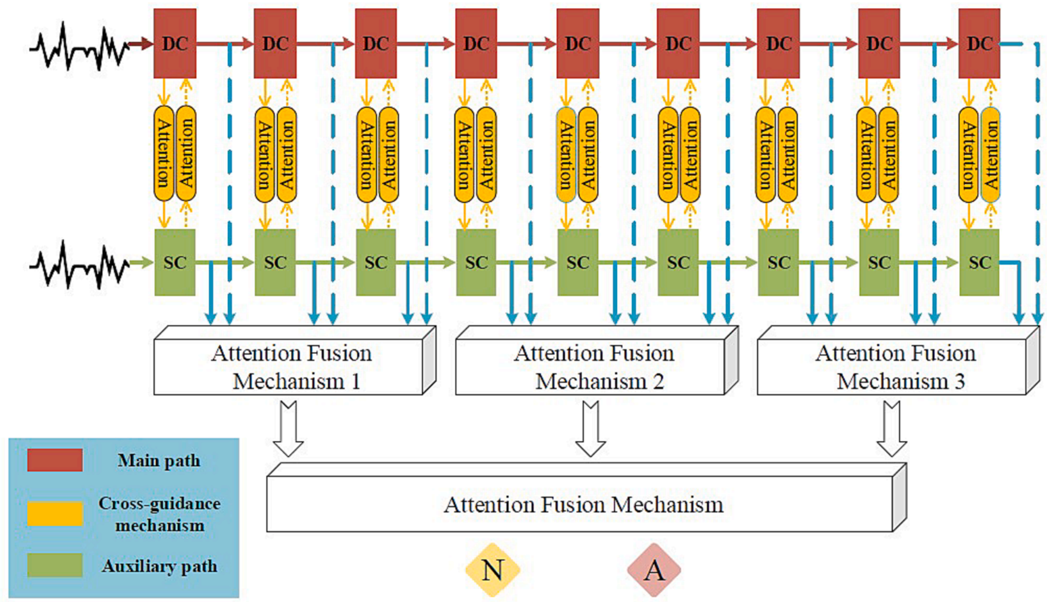

Fig. 2. The structure of LCG-Net. SC represents standard convolution, DC represents depth separable convolution, N represents non-atrial fibrillation and AF represents atrial fibrillation.

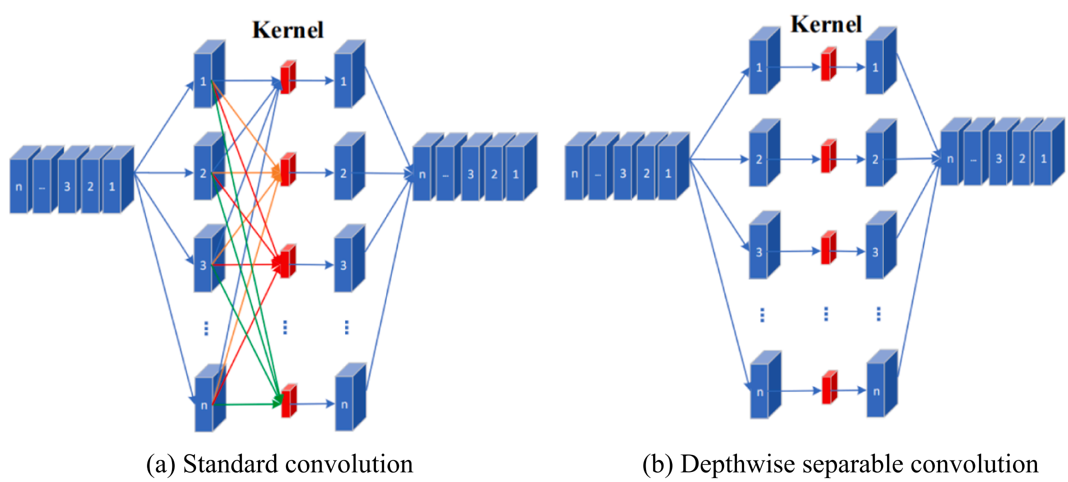

Fig. 3. Schematic diagram of standard convolution and depth-separable convolution.

a close temporal relationship. Purely using DC to extract features cuts off information exchange between different channels, which destroys the continuity and integrity of ECG information. Therefore, in this study, a standard convolution composed of 16 convolution kernels was used to establish an auxiliary path. The characteristic of full coverage of all channels by standard convolution was utilized to retain the information coupling relationship between different convolution channels.

2.2.2. Layer-by-layer cross-guidance mechanism

To address the issue of information flow obstruction between the main and auxiliary paths in the distributed network, this study improved the cross-guidance mechanism in image segmentation [36] and generalized it to the interaction between each layer of convolution in LCG-Net, achieving information exchange between layers of different paths. This study used lightweight attention module to replace the convolution and deconvolution layers in the cross-guidance mechanism [61]. Leveraging the characteristic of quickly identifying important information by attention mechanisms, the AF information at the same depth in different paths was encoded according to its importance and transmitted to the corresponding feature map in the other path through cross-multiplication. Enabling mutual guidance between the main and auxiliary paths and achieving mutual learning between them. When processing the feature maps output by different paths, this study used concatenation instead of element-wise addition. This reduced the inference cost further and achieved information transmission between the main and auxiliary paths. The feature extracted by the main path is compressed and weighted into the auxiliary path, and the feature extracted by the auxiliary path is compressed and weighted into the main path.

The attention module used in the cross-guidance mechanism layer by layer is in the form of Squeeze-and-Excitation (SE-Net) [37]. Compared with other modules, such as Convolutional Block Attention Module (CBAM) [38] and Selective Kernel Networks [39], SE-Net has a simpler structure and uses lightweight design ideas to automatically learn the importance of different channels. The inference process of the cross-guidance mechanism is shown in Eqs. (7)–(12).

First, the feature maps u_{c1} and u_{c2} from the main and auxiliary paths at the same layer are compressed into feature values z_{value1} and z_{value2} using the squeeze operation $F_{sq}(\bullet)$.

$$z_{value1} = F_{sq}(u_{c1}) = \frac{1}{H_1 \times W_1} \sum_{i=1}^{H_1} \sum_{j=1}^{W_1} u_{c1}(i,j) \tag{7}$$

$$z_{value2} = F_{sq}(u_{c2}) = \frac{1}{H_2 \times W_2} \sum_{i=1}^{H_2} \sum_{j=1}^{W_2} u_{c2}(i,j) \tag{8}$$

After applying the same operation to all feature maps within the same layer, we obtain two feature vectors: $Z_{vector1}$ consisting of main path feature values and $Z_{vector2}$ consisting of auxiliary path feature values. Then, we apply an activation function $F_{ex}()$ to the feature vectors to learn the importance of each feature channel, and obtain weight vectors s_1 for the main path and s_2 for the auxiliary path.

$$s_1 = F_{ex}(Z_{vector1}, W_1) = \sigma(g(Z_{vector1}, W_1)) = \sigma(W_{12}\delta(W_{11}Z_{vector1})) \tag{9}$$

$$s_2 = F_{ex}(Z_{vector2}, W_2) = \sigma(g(Z_{vector2}, W_2)) = \sigma(W_{22}\delta(W_{21}Z_{vector2})) \tag{10}$$

W_{11}、W_{12}、W_{21} and W_{22} are parameter matrices for different fully connected layers. $\sigma(\bullet)$ represents the Sigmoid function, and $\delta(\bullet)$ represents the ReLU function.

Finally, we add the weight values s_{c2} from the auxiliary path to the feature map u_{c1} in the main path to obtain the guided feature map y_1. We also add the weight values s_{c1} from the main path to the feature map u_{c2} in the auxiliary path to obtain the guided feature map y_2.

$$y_1 = F_{scale}(u_{c1}, s_{c2}) = s_{c2} \bullet u_{c1} \tag{11}$$

$$y_2 = F_{scale}(u_{c2}, s_{c1}) = s_{c1} \bullet u_{c2} \tag{12}$$

Apply 9 cross-multipliers between the main path and the auxiliary path at the same layer of convolution, that is, perform 9 operations according to Eqs. (7)–(12), to achieve a cross-guided mechanism layer by layer. Specifically, we convert the output of the first layer of depth-wise separable convolution in the main path into weights through the cross-guided mechanism, and weight them in the output of the first layer of convolution in the auxiliary path. Similarly, we convert the output of the first layer of convolution in the auxiliary path into weights through the cross-guided mechanism, and weight them in the output of the first layer of depth-wise separable convolution in the main path. The same operation is applied from the second layer to the ninth layer.

2.2.3. *Attentional fusion mechanism*

After being guided by a layer-by-layer cross-guidance mechanism, there exists a complementary relationship between the feature maps output by the main and auxiliary paths. However, there is also a certain degree of information redundancy, and simple linear addition or cascading operations on the two types of feature maps cannot fully explore and utilize their potential relationship [40]. This also cannot fully consider the importance of different features for AF recognition, and can increase computational complexity, leading to problems such as gradient disappearance or explosion. For example, Zhang et al. [41] concatenated ECG feature maps from 12 leads together to obtain complete information from multiple leads for arrhythmia detection. However, through feature visualization techniques, it was found that there was overlap between the features from the 12 leads, and some irrelevant features were included. Li et al. [42] used a simple weighted fusion strategy for feature maps from different ECG encoding images, ignoring the detailed features between different images and blurring the edge information of the images to some extent. Cui et al. [43] simply concatenated the features extracted by the DL model with manually extracted features to improve accuracy by increasing the variety of features, but they did not consider the problem of overlapping information between DL features and manually extracted features.

ResNet [44] has demonstrated that deep layers can effectively utilize features extracted by shallow layers to enhance network performance, while also avoiding the issues of gradient vanishing and exploding. However, the feature maps of shallow layers in DL models are redundant [45], and directly adding them linearly to the feature maps of deep layers will increase the complexity of network learning. Variants of ResNet [46] have shown that adding shallow feature maps indiscriminately to deep feature maps can lead to a decrease in model performance due to information redundancy.

To achieve more refined processing of different features and implement accurate fusion strategies, this study proposes an attention fusion mechanism, as shown in Fig. 4. Here, BI represents the information from the main path, AI represents the information from the auxiliary path, and GAP represents global average pooling. The attention mechanism searches for features from different paths and learns to transform all feature maps into corresponding feature vectors. Then, the feature vectors between different layers are concatenated vertically to obtain a set of two-dimensional matrices. Finally, based on a two-dimensional convolutional template, the obtained two-dimensional feature matrices are convolved to obtain a one-dimensional feature vector for classification.

The inference formulas for the attention fusion mechanism are shown in Eqs. (13)–(16). Assuming that the feature maps from the main and auxiliary paths are concatenated, the output of the i-th layer of LCG-Net is $\mathrm{X} = \{X_1, X_2, X_3, \cdots, X_k\}$ with k feature maps, the output of the $(i+1)$-th layer is $\mathrm{Y} = \{Y_1, Y_2, Y_3, \cdots, Y_s\}$ with s feature maps, and the output of the $(i+2)$-th layer is $\mathrm{Z} = \{Z_1, Z_2, Z_3, \cdots, Z_l\}$ with l feature maps. First, the feature maps of each layer are compressed into feature values using global average pooling, and the feature values of each layer form the feature vector of that layer. Thus, the feature vectors of the i-th, $(i+1)$-th, and $(i+1)$-th layers are X, Y, and Z, respectively. Then, the feature vectors of different layers are preliminarily mapped using the attention mechanism, as shown in Eqs. (13)–(15).

$$f(X) = \sigma(W_1 * \mathrm{X} + B_1) \tag{13}$$

$$f(Y) = \sigma(W_2 * \mathrm{Y} + B_2) \tag{14}$$

$$f(Z) = \sigma(W_3 * \mathrm{Z} + B_3) \tag{15}$$

W_1、W_2 and W_3 represent the weight matrix, B_1、B_2 and B_3 represent the deviation coefficients, and $\sigma(\hat{\mathrm{A}}\cdot)$ represents the Sigmoid activation function.

Assuming that after mapping, three feature vectors $x = \{x_1, x_2, x_3, \cdots, x_t\}$、$y = \{y_1, y_2, y_3, \cdots, y_t\}$ and $z = \{z_1, z_2, z_3, \cdots, z_t\}$ of length t are obtained, a vertical concatenation strategy is applied to the mapped feature vectors to obtain a $3 \times t$ two-dimensional matrix. Each column of the matrix is convolved using a two-dimensional convolutional template, and the feature values from different layers are weighted and fused according to learned weights to obtain the fused feature vector BAI. The fusion process is shown in the following equation:

$$BAI = \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_t \\ y_1 & y_2 & y_3 & \cdots & y_t \\ z_1 & z_2 & z_3 & \cdots & z_t \end{bmatrix} * \begin{bmatrix} w_{t1} \\ w_{t2} \\ w_{t3} \end{bmatrix} = \{bai_1 \quad bai_2 \quad bai_3 \quad \cdots \quad bai_t\} \tag{16}$$

where the eigenvalues of the fused feature vector BAI $bai_i = w_{t1} \times x_i + w_{t2} \times y_i + w_{t3} \times z_i$; w_{t1}, w_{t2} and w_{t3} are the weights obtained from the two-dimensional convolution operator learning.

3. Results

3.1. *Evaluation indicators*

The performance of the proposed method in this study is evaluated based on parametric quantity (P), calculated quantity (F), overall accuracy (Acc) and F1 score. The calculation process for the two indicators is as follows:

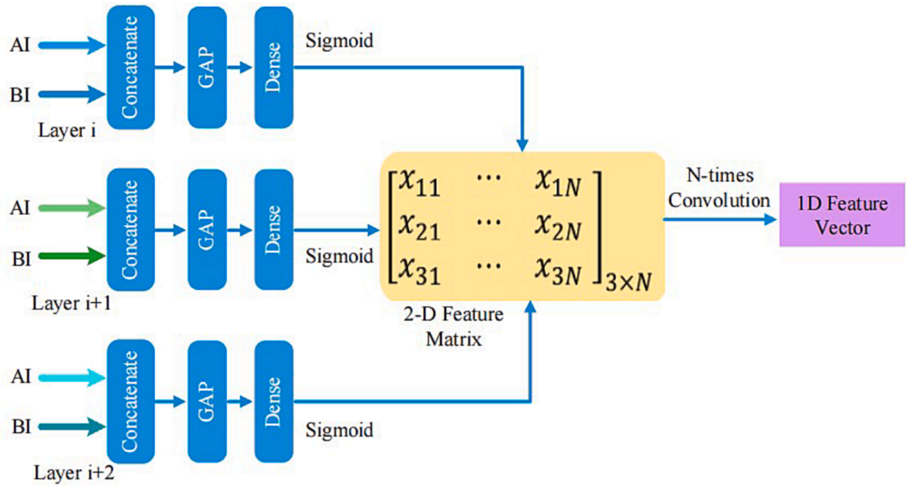

Fig. 4. Schematic diagram of the attention fusion mechanism. BI represents the information from the main path, AI represents the information from the auxiliary path, and GAP represents global average pooling.

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

$$F1 = 2 \times \frac{Pre \times \text{Recall}}{Pre + Recall} \tag{18}$$

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

$$Pre = \frac{TP}{TP + FP} \tag{20}$$

TP represents true positive (AF is correctly classified as AF), TN represents true negative (normal is correctly classified as normal), FN represents false negative (AF is incorrectly classified as normal), and FP represents false positive (normal is incorrectly classified as AF).

3.2. Results of LCG-Net on clinical dataset

The training process of this study used an early stopping strategy, where training would stop if the Acc of the validation set did not improve for five consecutive epochs. The results of LCG-Net on SPHDB are shown in Table 1, where the average Acc and average F1 score of LCG-Net reached 96.07 % and 96.01 %, respectively. This indicates that the features extracted by LCG-Net can effectively distinguish AF samples from N samples, with low false positive and false negative rates in AF identification. Due to the randomness of DL and sample distribution, the results of LCG-Net on different test sets fluctuated by 1 % to 8 %. However, the lowest average Acc and average F1 score of LCG-Net still reached 94.88 % and 94.25 %, respectively. This indicates that the LCG-Net proposed in this study exhibits good generalization in identifying AF.

3.3. Result analysis and improvement

In the experiment, we found that LCG-Net has serious false positives for continuous segments of individual patients. When G2 and G3 are used as the test sets, their false positive rates are higher than when G1 is used as the test set. To explore the reasons for misclassification, we selected misclassified samples for analysis. Fig. 5 illustrates some of the samples, which are false positives. Through analysis, we found that in this part of the samples, the RR interval was regular, but there was generally unstable baseline. LCG-Net classified the unstable baseline as disordered fibrillation and considered the baseline disappearance as an important indicator for clinical doctors to judge AF. Therefore, LCG-Net mistakenly identified it as AF. For example, in samples (a) and (c) of Fig. 5, the amplitude of the P wave and T wave were similar, and the baseline disappeared after the T wave. However, because of the regular RR interval and the presence of P wave, clinical doctors determined that the segment was N. In samples (b) and (d) of Fig. 5, the ECG signal showed frequent tremors, indicating that there was significant noise interference or other diseases mixed in (Fig. 5(d) is atrial flutter mixed with noise), which caused LCG-Net to misdiagnose N as AF. In order to reduce the false positive rate of LCG-Net in the above-mentioned scenario, this study retrained LCG-Net using SPHDB Plus. The results of LCG-Net on SPHDB Plus are shown in Table 2. After data augmentation, the overall performance of LCG-Net on different training sets and different test sets has been improved.

- The overall average Acc and F1 score were improved by 0.45 % and 0.52 %, respectively.
- When the training set was different and the test set was the same, the highest improvement in average Acc and F1 score was 0.97 % and 1.57 %, respectively.

Table 1
Acc (%) and F1 score (%) of LCG-Net on SPHDB.

Test / Train	G1		G2		G3		G4		G5		Mean	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
G1	–	–	94.70	94.92	94.99	95.19	95.35	95.46	97.59	97.63	95.66	95.80
G2	97.26	97.26	–	–	95.67	95.80	94.74	94.91	98.12	98.11	96.44	96.52
G3	97.80	97.82	97.33	94.71	–	–	94.27	94.52	99.16	99.16	97.14	96.55
G4	98.09	98.09	90.70	90.48	91.12	91.20	–	–	98.15	98.24	94.52	94.50
G5	97.84	97.86	96.80	96.87	96.13	96.26	95.62	95.76	–	–	96.60	96.69
Mean	97.75	97.76	94.88	94.25	94.48	94.61	94.99	95.16	98.26	98.29	**96.07**	**96.01**

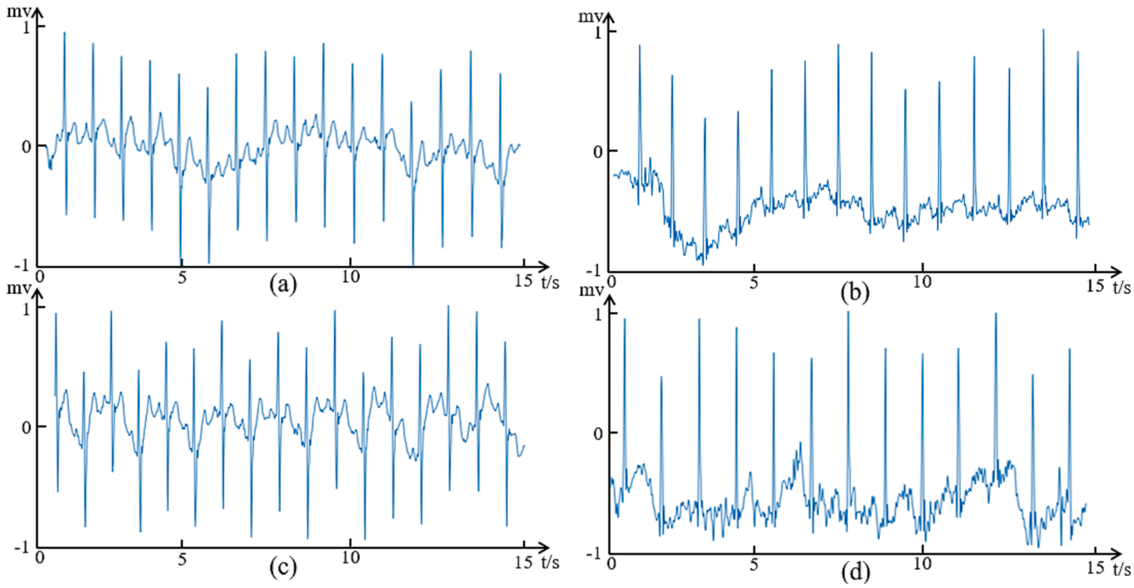

Fig. 5. Examples of false positive ECG segment.

Table 2
Acc (%) and F1 score (%) of LCG-Net on SPHDB Plus.

TTest / TTrain	G1		G2		G3		G4		G5		Mean	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
G1 + AD	–	–	94.49	94.46	94.99	95.06	95.18	95.21	98.71	98.71	**95.84**	**95.86**
G2 + AD	97.08	97.04	–	–	96.00	96.02	95.38	95.41	98.13	98.12	**96.65**	**96.65**
G3 + AD	98.17	98.17	97.38	97.39	–	–	95.53	95.61	98.97	98.96	**97.51**	**97.53**
G4 + AD	97.85	97.85	95.56	95.56	94.39	94.30	–	–	97.74	97.71	**96.39**	**96.36**
G5 + AD	97.79	97.80	95.98	95.99	95.41	95.43	95.73	95.82	–	–	96.23	96.26
Mean	97.72	97.72	**95.85**	**95.85**	**95.20**	**95.20**	**95.46**	**95.51**	**98.39**	**98.38**	**96.52**	**96.53**

- When the test set was different and the training set was the same, the highest improvement in average Acc and F1 score was 1.87 % and 1.86 %, respectively.
- The lowest Acc and F1 score in SPHDB (Table 1) (G4 training set, G2 test set) were improved by 4.86 % and 5.08 %, respectively.

This indicates that after data augmentation, LCG-Net not only retained the AF features learned in SPHDB, but also learned the difference between unstable baseline and baseline disappearance caused by AF based on the newly added patients. This enables LCG-Net to make accurate judgments for rare and unconventional ECGs while ensuring good clinical versatility.

4. Discussion

4.1. Comparison with advanced research

To demonstrate the superiority of LCG-Net in the AF detection task, this study compared it with relevant literature in recent years that used AFDB for AF detection. The results are shown in Table 3. Our proposed LCG-Net achieved an accuracy of 99.66 %, specificity of 99.18 %, and sensitivity of 99.88 %. It's worth noting that in comparison to the study [47], our method's sensitivity is only 0.05 % lower, but specificity is higher by 2.15 %, and accuracy is higher by 3.07 %. This indicates that our proposed approach outperforms existing algorithms in the recognition of AF. Compared to the literature in Table 3, LCG-Net utilizes a distributed network architecture when extracting information about AF. It employs multiple convolutional approaches to learn AF information from ECG data from various angles. It achieves information interaction between different paths through a layer-by-layer cross-guidance mechanism, promoting mutual assistance between the main path and auxiliary paths. The attention fusion mechanism is used to merge features from different layers and different paths. Weight values for each feature are objectively assigned based on the loss function through back-propagation, enhancing the role of every detail in AF detection.

Table 3
Comparative results of AFDB.

Author	Year	Method	Acc (%)	Spe (%)	Sen (%)
Ahmad et al. [54]	2020	HAN	99.08	98.54	99.08
Dang et al. [47]	2021	Bi-LSTM	96.59	97.03	**99.93**
Radhakrishnan et al. [48]	2021	AFCNN	98.92	98.94	98.91
Wang et al. [49]	2021	Bi-GRU	98.30	98.20	98.40
Subramanyan et al. [50]	2022	CNN + RNN	99.16	98.64	99.16
Wang et al. [51]	2022	IENN	97.90	97.80	98.00
Feng et al. [52]	2022	IB-LSTM	98.20	98.00	98.40
Ramkumar et al. [53]	2022	CNN-BAROA	98.00	89.00	93.00
Ours	2023	LCG-Net	**99.66**	**99.18**	99.88

4.2. Comparison of lightweight models

To further demonstrate the superiority of LCG-Net, this study designed a set of control experiments using MT-DCNN [15], Squeeze Net [28], Mobile Net [33], 1D-DCNN [54], and Ghost Net [55] as reference models. The test results of the reference models on SPHDB Plus are shown in Table 4, and the computational cost comparison is shown in Table 5. The results show that LCG-Net achieved an average Acc of 98.39 % and an average F1 score of 98.38 %, with only 39.04 K parameters and 8.16 M computations.

MT-DCNN used the idea of multi-task mutual assistance and achieved a maximum Acc and F1 score of 95.46 % and 95.37 % on different datasets. But according to Table 5, the parameter and computational cost of LCG-Net are only 0.3 % and 0.5 % of MT-DCNN, respectively, and the computational cost of LCG-Net is much lower than that of MT-DCNN.

Squeeze Net, as one of the earliest recognized lightweight models, completely failed when directly applied to this study, indicating that there are still significant problems with the generalization of DL models across different domains.

Mobile Net, as a mature lightweight model, achieved a maximum Acc and F1 score of 98.64 %. But as shown in Table 5, the computational cost and parameter of LCG-Net are only 1 % of Mobile Net, and its complexity is much lower than that of Mobile Net, which is more in line with the requirements of mobile deployment.

1D-DCNN is a lightweight model based on the inverted residual module, and the computational cost is only higher than that of LX-Net. However, it failed when facing training and test sets with significantly different sample distributions. Only when G3 was used as the training set, did 1D-DCNN perform well, achieving an Acc and F1 score of 98.75 % and 98.74 %, but its lightweight degree, stability, and learning ability are inferior to LCG-Net.

Ghost Net is a new generation of lightweight models proposed by Huawei Noah's Ark Laboratory, which achieved an Acc and F1 score of 98.23 % and 98.24 %, respectively, on SPHDB Plus. But as shown in Table 5, the parameter and computational cost of LCG-Net are less than 5 % of Ghost Net, indicating that LCG-Net is superior to Ghost Net in both performance and computational cost.

In summary, the LCG-Net proposed in this study not only has high Acc, low false positive rate, and low false negative rate in AF recognition tasks, but also has better overall computational cost and stability than existing lightweight models.

4.3. T-SNE scatter plot

To demonstrate that LCG-Net can select different features of N and AF segments, we randomly selected 10,000 samples (5,000 N samples and 5,000 AF samples) from the test set as the visualization dataset. Visualize the features output by LCG-Net's attention fusion mechanisms 1, 2, and 3 using the t-SNE algorithm [56]. The visualization results are shown in Fig. 6, where Fig. 6(a) shows the feature distribution of the first three layers, Fig. 6(b) shows the feature distribution of the middle three layers, and Fig. 6(c) shows the feature distribution of the last three layers. In the first three layers, AF features and N features are interwoven and overlapping, and the two types of features are distributed in one cluster without a clear boundary. In the middle three layers, the geometric distance between the two types of features increases significantly, but the boundary is still blurred, and there is a cluster of N features distributed in the AF feature cluster in the upper-middle part of Fig. 6(b). In the last three layers, the two types of features are completely affine to two clusters, and the geometric distance further increases, and the boundary between the two clusters is obvious, and there is no obvious intersection between the two types of features. Through visualization techniques, it can be intuitively seen that LCG-Net can accurately extract the features of AF and N after deep learning.

Table 5
P (K, Kilobyte) and F (M, Mega) of the comparative model on SPHDB Plus.

	LCG-Net	1D-DCNN	MT-DCNN	Mobile Net	Squeeze Net	Ghost Net
P	**39.04**	145.38	9089.8	3211.33	366.27	830.78
F	**8.16**	122	181.05	821.15	255.19	257.85

4.4. Advantages of this study

As DL models continue to evolve, the demand for massive data and computing power will continue to increase, and traditional models are difficult to meet real-time requirements [57]. Therefore, achieving lightweight models and reducing the requirements of models for computing power has become a trend for future development. In this study, we propose a LCG-Net for AF recognition. Compared with DL models previously used for AF detection [58,59], LCG-Net further simplifies the complexity of the algorithm while ensuring high-precision AF recognition, realizing the design concept of lightweight. Compared with lightweight AF detection algorithms [60], LCG-Net not only pursues lightweight while taking into account the model's feature mining ability, but also improves the reliability and accuracy of the algorithm.

Considering that DC blocks the information transmission between different channels while reducing the computational complexity in the feature extraction process, which reduces the network's ability to extract AF features to some extent [34], this study uses standard convolution to assist DC in the feature extraction process. Each layer in LCG-Net is similar to replacing the 1 × 1 convolution in the Expand layer of Squeeze Net with DC. According to Eqs. (5) and (6), when performing convolution calculations on the same sample, the parameter amount of DC is K_D/C times that of 1 × 1 convolution, and the computational complexity of DC is $K_D \times (L - K_D + 1)/(C \times L)$ times that of 1 × 1 convolution. In the network design process, the size of the convolution kernel K_D is greater than or equal to 3 and much smaller than the number of convolution kernels C, so the parameter amount and computational complexity of DC are much smaller than those of 1 × 1 convolution. Moreover, the receptive field of DC is larger than that of 1 × 1 convolution, and it captures a larger range of features in the feature extraction process. Compared with the Expand layer, the local feature extraction ability of each layer in this study has been improved, and the degree of lightweight has been further improved.

To achieve information transmission between layers in a distributed network, this research proposes a layer-by-layer cross-guidance mechanism, which utilizes the feature of lightweight attention modules to quickly capture key information and guide the information flow of the same layer convolution in different paths. The main path accurately captures important local information of AF and encodes the information obtained through attention mechanism to the auxiliary path, enabling the auxiliary path to effectively obtain the missing information of the

Table 4
Acc (%) and F1 score (%) of the comparative model on SPHDB Plus.

Model	G1		G2		G3		G4		G5	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
MT-DCNN	95.28	95.31	93.39	93.48	91.81	91.87	92.06	92.34	95.46	95.37
Mobile Net	97.74	97.76	95.22	95.31	95.57	95.61	95.16	95.28	98.64	98.64
Ghost Net	97.52	97.53	94.76	94.84	95.43	95.42	94.72	94.87	98.23	98.24
Squeeze Net	Gradient disappearance or explosion									
1D-DCNN	97.80	97.82	97.33	94.71	–	–	94.27	94.52	99.16	99.16
LCG-Net	97.72	97.72	95.85	95.85	95.20	95.20	95.46	95.51	98.39	98.38

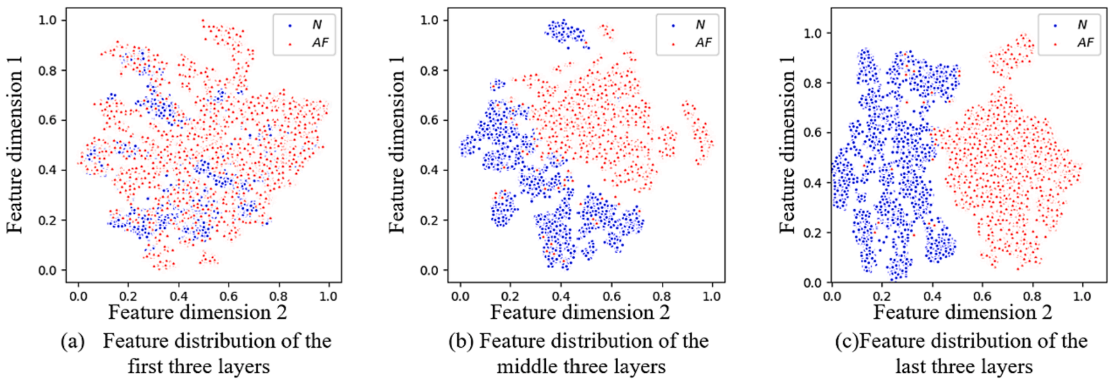

Fig. 6. The evolution process of N-class features and AF features.

main path. Based on the feature of standard convolution channel full coverage, the auxiliary path focuses on learning and supplementing the missing information of the main path, and encodes the extracted information through attention mechanism, and weights it into the feature map extracted by the main path. After applying the layer-by-layer cross-guidance mechanism, LCG-Net enhances the information interaction between different paths, not only extracting the depth information of AF, but also maintaining the inherent relationship between different information, achieving functional complementarity between different paths.

To achieve feature fusion between different paths and layers, this research utilizes attention mechanism and two-dimensional convolution templates to design an attention fusion mechanism for feature selection and fusion. Firstly, attention mechanism is used to preliminarily screen all features and capture the features with high relevance to AF recognition. Then, two-dimensional convolution templates are used to fuse features from different layers, leveraging their wide receptive field and strong learning ability. Compared with the original feature set, the feature set after attention mechanism screening has reduced information redundancy and shortened the geometric distance between the predicted probability of LCG-Net, making it easier to learn the fusion weights. Unlike simple concatenation or linear addition, attention fusion mechanism uses convolution calculation to fuse features, weighting the feature values from different layers and amplifying the role of deep-level features in AF recognition. Weighting each feature value rather than the entire feature vector emphasizes the role of detailed information. Unlike manually set weight values, the weight values of attention fusion mechanism are obtained through reverse propagation of the loss function, which is more objective. After applying the attention fusion mechanism, not only the AF information from different paths and layers is accurately fused, but also the common problem of information redundancy in feature fusion is eliminated.

In addition, we collected data from 252 AF patients to establish a dataset for AF detection. Compared with the well-known AFDB, the SPHDB Plus dataset established in this study is more diverse in terms of personnel composition and age range, and the number of patients and recording time exceed that of AFDB. Therefore, validation of the model on SPHDB Plus is more clinically relevant. In terms of data partitioning, unlike K-fold cross-validation, the sample size of the training set in this study is much smaller than that of the test set, which is consistent with the actual situation in clinical practice where labeled samples are far fewer than unlabeled samples. Therefore, the experimental results obtained in this study are more authentic. In terms of testing the effectiveness of the algorithm, unlike most studies, this study simultaneously tested the robustness of the model when faced with different training sets and the generalization when faced with different test sets.

4.5. Limitations of this study

Firstly, according to the analysis of clinical doctors, our model performs well for low-amplitude signals, but the performance is reduced for signals with large amplitude differences or unstable baselines. Secondly, the focus of our research is on algorithm development, and hardware deployment has not been implemented. In future work, our team will use traditional machine learning algorithms to calibrate deep learning models in a targeted manner, further explore the stability of lightweight algorithms, and deploy lightweight algorithms based on FPGA.

5. Conclusion

This study proposes a LCG-Net, and LCG-Net utilizes a distributed approach to construct two lightweight paths. While using depth-wise separable convolution to extract AF deep-level information, a small amount of standard convolution is used to compensate for the problem of information interaction between different channels of depth-wise separable convolution. The layer-by-layer cross-guidance mechanism guides the interaction of information between the main and auxiliary paths through attention encoding. The attention fusion mechanism accurately selects and objectively fuses each feature value from different paths and layers. Compared with existing algorithms and classic lightweight DL models, LCG-Net achieves more accurate and stable AF detection, reduces algorithm costs and hardware requirements, and has good clinical application prospects. The experimental results are of great significance to guide the analysis of various physiological signals, such as EEG signals, heart sounds, and pulse signals, as well as the research in the field of computer vision, such as image recognition and segmentation. At the same time, this paper establishes a clinical dataset for AF analysis for other scholars to use.

Funding

This work was supported by the National Natural Science Foundation of China [grant numbers 82072014], the National Key R&D Program of China [grant numbers 2019YFE010670], the Shandong Province Natural Science Foundation [grant numbers ZR2020MF028].

Code and data

The code is available at https://github.com/LiYongjian206/LX-Net.
The SPHDB is available at https://data.mendeley.com/datasets/g23n565xh3/1.

CRediT authorship contribution statement

Yongjian Li: Conceptualization, Methodology, Writing – original draft. **Meng Chen:** Data curation. **Xing'e Jiang:** Visualization. **Lei Liu:**

Supervision. **Baokun Han:** Validation. **Liting Zhang:** Investigation. **Shoushui Wei:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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